

COMPUTATIONAL APPROACHES TO LEXICAL SEMANTIC CHANGE DETECTION WITH WIC-BASED AND LLM-BASED SENSE INDUCTION

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PhD Thesis Defense

Outline

- 1 Introduction
- 2 Study 1
- 3 Study 2
- 4 Study 3

Introduction

- Language is constantly evolving, reflecting processes within both language and society.

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Lexical Semantic Change Detection (LSCD)

LSCD is the computational task of automatically detecting changes in word meanings over time.

Why is LSCD important?

Applications of LSCD

- **Linguistics**

Reveals how social and cultural dynamics, cognitive processes, and lexical structure interact to shape word usage over time.

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- **Linguistics**

Reveals how social and cultural dynamics, cognitive processes, and lexical structure interact to shape word usage over time.

- **Lexicography**

Tracks the emergence of new word senses and helps refine historical dictionaries.

(Schlechtweg, 2023)

Applications of LSCD

- **Social and Political Science**

Investigates cultural and ideological shifts, the evolution of political discourse, and semantic signals tied to societal phenomena such as armed conflict

(Karjus and Cuskley, 2023; Kutuzov et al., 2019)

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(Karjus and Cuskley, 2023; Kutuzov et al., 2019)

- **Digital Humanities**

Enables computational analysis of historical corpora across disciplines

(Schlechtweg, 2023)

Thesis Perspective

This thesis studies LSCD from a sense-oriented perspective

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Across three studies, it examines how sense-level information can be used to:

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Across three studies, it examines how sense-level information can be used to:

- Build a reliable Spanish LSCD resource
- Improve semantic change detection by optimizing the underlying **Word Sense Induction (WSI)** task
- Evaluate how **Large Language Models (LLMs)** judge semantic proximity between two usages

Thesis Roadmap

① LSCD Foundations

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- ② Study 1 — Building a Spanish LSCD Dataset

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- ⑤ Conclusions

Lexical Semantic Change Detection

- **Binary Task**

Given a set of target words, determine whether each word has undergone semantic change between corpora C_1 and C_2 .

- **Ranking Task**

Rank target words according to the degree of semantic change between C_1 and C_2 by comparing their sense frequency distributions.

(Schlechtweg et al., 2020)

Lexical Semantic Change Detection

COMPARE Task

Given two usages of the same target word:

$$u_1 \in C_1 \quad \longleftrightarrow \quad u_2 \in C_2$$

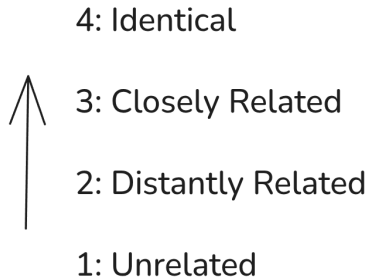


$$\text{sim}(u_1, u_2)$$

Lexical Semantic Change Detection

How do humans annotate semantic proximity samples?

Human Annotation of Semantic Proximity



(Schlechtweg et al., 2018)

Example

- **Usage A (1850):**
The prisoner was confined to a narrow *cell* in the tower.
- **Usage B (2001):**
She forgot her *cell* at home and missed the call.
- **Judgment:** 1 – Unrelated

Word Usage Graph (WUG)

$$G = (U, E, w)$$

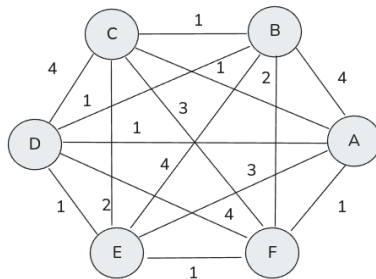
- $u \in U$: represents a **word usage**
- $(u_1, u_2) \in E$: represents a **pair of usages**
- $w(u_1, u_2)$: represents their **semantic proximity**

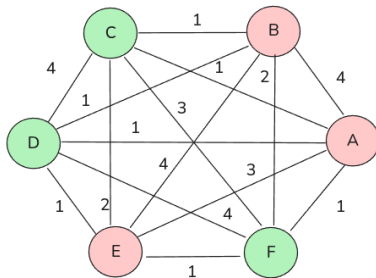
WUG: a **weighted, undirected graph** of word usages

(Schlechtweg et al., 2020)

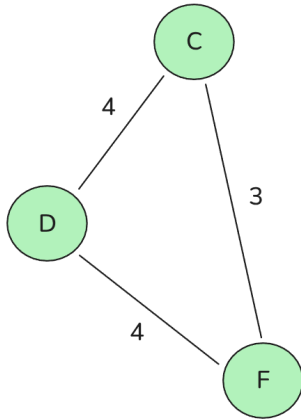
Example

ID	Period	Usage	Meaning
A	1810	Fui al banco a retirar dinero.	financial institution
B	1855	El banco me concedió un préstamo.	financial institution
E	1906	El banco central emitió un comunicado oficial.	financial institution
C	1994	Nos sentamos en el banco del parque a descansar.	bench
D	2008	El banco estaba mojado después de la lluvia.	bench
F	2020	Se quedó dormido en un banco de madera junto al río.	wooden bench

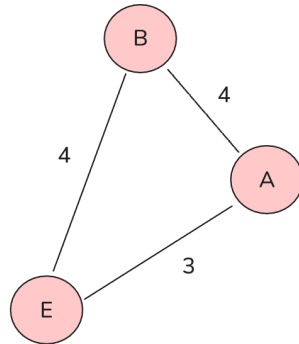




Sense Frequency Distribution



$$\text{SFD} = (2, 0)$$



$$\text{SFD} = (0, 3)$$

From SFDs to Graded Semantic Change - Jensen-Shannon Distance

Jensen-Shannon Distance (JSD)

Measures the degree of semantic change between two sense frequency distributions.

$$\text{SFD}_1 = (3, 0)$$

$$\text{SFD}_2 = (0, 3)$$



$$\boxed{\text{JSD}(\text{SFD}_1, \text{SFD}_2) = 1.0}$$

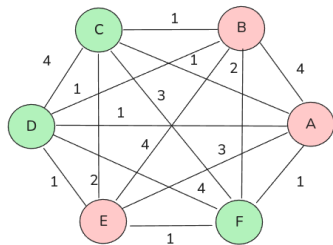
Maximum semantic change

(Schlechtweg, 2023, p. 38; Lin, 1991)

From SFDs to Graded Semantic Change - Average Pairwise Distance

Average Pairwise Distance (APD)

(Kutuzov and Giulianelli, 2020)



$$\text{APD}(w) = \frac{1}{|U_1||U_2|} \sum_{u_1 \in U_1} \sum_{u_2 \in U_2} d(u_1, u_2)$$

Average semantic distance between usages
from the two periods.

SemEval-2020 Task 1: A Benchmark for LSCD

- First large-scale shared task for LSCD
- Standardized evaluation framework
- Introduced two subtasks:
 - Binary classification
 - Ranking
- Provided manually annotated datasets for English, German, Latin, and Swedish
- The benchmark used in this thesis builds on **Word Usage Graphs (WUGs)**

(Schlechtweg et al., 2020)

SemEval-2020 Task 1

Unsupervised LSCD

Binary Task

Detect whether a word has undergone semantic change

Accuracy

Ranking Task

Rank words according to their degree of semantic change

Spearman rank correlation

(Schlechtweg et al., 2020)

SemEval-2020 Task 1

Four datasets and a set of target words

$$W = \{w_1, w_2, \dots, w_n\}$$

Language	C_1	C_2	n
English	1810–1860	1960–2010	37
German	1800–1899	1946–1990	48
Latin	–200–0	0–2000	40
Swedish	1790–1830	1895–1903	31

(Schlechtweg et al., 2020)

SemEval-2020 Task 1

Main Finding

Type-based models outperformed token-based models, despite the theoretical advantage of contextualized representations for capturing the different senses of a word.

System type	Subtask 1	Subtask 2
Type-based systems	.625	.329
Token-based systems	.598	.258

(Schlechtweg et al., 2020)

Benchmarks Beyond SemEval-2020 Task 1

Italian Dataset

- Binary task
- Type-based systems dominated

Russian Dataset

- Three time periods
- **COMPARE** task
- Token-based systems outperformed type-based systems

Research Gap

Although recent LSCD methods have achieved strong performance,
it remains unclear how **sense-level information**
can best be exploited for lexical semantic change detection.

An Open Challenge

- **Inconsistent findings across languages**
 - Italian and Russian datasets used different annotation approaches
- **No WUG-based benchmark beyond English and German**
 - No benchmark using the same WUG methodology existed to test whether the SemEval findings generalize to other languages

Extending LSCD to Spanish

- **A major world language**

More than 450 million native speakers across more than 20 countries

- **Underrepresented in NLP benchmarks**

Spanish remains comparatively underrepresented in widely used NLP resources and benchmarks

- **Rich lexical polysemy**

The high degree of polysemy makes Spanish a particularly interesting language for studying lexical semantic change

Research Questions

- Can we create a reliable Spanish dataset for the LSCD field?
- Can LSCD systems developed for other languages maintain their performance when applied to Spanish?

DWUG ES – LSCD Dataset

- **Two diachronic corpora**

From the OPUS project (Tiedemann, 2012)

Corpus	Periods	Tokens
Old	1810–1906	~13M
Modern	1994–2020	~22M

DWUG ES – LSCD Dataset

- **Large-scale LSCD dataset**
~62K judgments collected from 12 native Spanish speakers
- **100 annotated target words**

Annotation Process and Agreement

- **12 native Spanish speakers**
4 Chileans, 4 Colombians, 2 Cubans, 1 Spaniard, 1 Venezuelan

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40 usages in total per target word

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- **WUG annotation using the DUREl 4-point scale**
- **From annotations to gold labels**
Correlation clustering → Sense Frequency Distributions → Jensen–Shannon Distance → gold labels

Annotation Process and Agreement

Partition	Words	Krippendorff's α
Development	20	.53
Evaluation	60	.58
Discarded	20	.27
Full	100	.53

Krippendorff's α measures inter-annotator agreement.

Annotator Disagreement by Nationality

- **Modern usage (C_2)**

“La investigación ha demostrado que las plantas de maíz . . .”

Meaning: biological organism (crop plants)

- **Historical usage (C_1)**

“nada se tocaría a la planta y ornamentación de la fábrica árabe . . .”

Meaning: architectural layout

Colombian	Spaniard	Cuban	Chilean
1	4	4	3

Annotator judgments for the pair of usages

LSCDiscovery: A Shared Task in Spanish LSCD

LSCDiscovery

A shared task was organized with two mandatory phases:

Phase	Task	Description	Metric
Phase 1	Graded Change Discovery	Rank all vocabulary words (4,385)	Spearman's ρ
Phase 2	Binary Change Detection	Classify pre-selected words as changed / not changed	F1

(Zamora-Reina et al., 2022)

LSCDiscovery: A Shared Task in Spanish LSCD

Optional Tasks

Task	Description
Graded Change Detection	Same as Discovery, but on pre-selected annotated words
COMPARE	Predict the negated DUREl score
Sense Gain Detection	Identify only words that gained senses
Sense Loss Detection	Identify only words that lost senses

(Zamora-Reina et al., 2022)

LSCDiscovery: A Shared Task in Spanish LSCD

- **Main characteristics**

- Fully unsupervised
- 6 teams participated in Phase 1
- 7 teams participated in Phase 2

- **Novelty**

- Introduced the **Discovery scenario**
- Required systems to cover the **full vocabulary**, as lexicographers would need to do in practice

(Zamora-Reina et al., 2022)

LSCDiscovery: A Shared Task in Spanish LSCD

Results – Phase 1

Rank	System	Representation	Approach	Spearman's ρ
1	GlossReader	Token (XLM-R)	APD	.735
2	DeepMistake	Token (XLM-R)	APD	.702
3	HSE	Token (BERT)	Clustering	.553
–	Baseline 1	Type (SGNS)	Cosine Distance	.543

Usage-based approaches dominated the top positions,
while **clustering** also achieved competitive performance.

(Zamora-Reina et al., 2022)

LSCDiscovery: A Shared Task in Spanish LSCD

Key Findings and Conclusions

- ① **Token-based systems dominated**
 - The strongest systems relied on contextualized token representations and APD.
- ② **Performance for graded change was comparable to previous shared tasks**
- ③ **Sense-level approaches remained competitive**
 - Clustering-based systems ranked among the stronger approaches.

(Zamora-Reina et al., 2022)

LSCDiscovery: A Shared Task in Spanish LSCD

Key Findings and Conclusions

- ④ APD has a clear upper bound
- ⑤ COMPARE closely approximates the Graded Change Task

Measure	Correlation with JSD
Gold COMPARE	.92

(Schlechtweg, 2023, p. 66; Zamora-Reina et al., 2022)

LSCDiscovery: A Shared Task in Spanish LSCD

Key Findings and Conclusions

- ⑥ **Clustering methods were among the best-performing systems**
 - Current approaches exploit correlations between change measures, they do not explicitly model the annotation process

(Zamora-Reina et al., 2022)

LSCDiscovery – Insights

- Aggregation-based approaches are approaching a performance ceiling
- Clustering-based systems emerged among the top-performing approaches
 - Suggesting that explicitly modeling the **sense-level structure** underlying the annotation process may provide advantages for LSCD
- The relationship between WSI and LSCD performance remained unexplored

From Resource to Modeling

- LSCDiscovery showed that sense-level approaches can be competitive
 - Clustering-based systems appeared among the top-performing approaches
- However, the role of WSI quality in LSCD performance remained unclear

Research Questions

- 1 Does optimization for WSI transfer to LSCD?
- 2 Does optimization for LSCD transfer to WSI?

Hypothesis

**Optimizing clustering configurations for WSI
will improve their downstream performance on LSCD.**

Annotation Procedure

Problem

- **Human annotation is expensive**
 - Annotators must judge pairs of usages for each target word
- **LSCDiscovery required $\sim 62K$ judgments for only 100 words**
- **Scaling to the full vocabulary is impractical**

Solution

Automate semantic proximity judgments using Word-in-Context (WiC) models

WiC Task

Given a target word w and two contexts c_1 and c_2 :

(w, c_1)

(w, c_2)

$$f(w, c_1, c_2) \rightarrow \begin{cases} 1 & \text{same meaning} \\ 0 & \text{different meaning} \end{cases}$$

Binary classification of semantic relatedness

(Pilehvar & Camacho-Collados, 2019)

WiC Models

$$\underbrace{(u_1, u_2)}_{\text{usage pair}} \xrightarrow{\text{WiC model}} \hat{y}(u_1, u_2)$$

$\hat{y}(u_1, u_2) \approx$ human semantic proximity judgment

- **Automate semantic proximity judgments**
 - Replace manual judgments with model predictions
- **Enable scalable annotation**
 - Predictions can be generated for large numbers of target words

WiC Model – DeepMistake

$$(u_1, u_2) \xrightarrow{\text{DeepMistake}} P(\text{same sense} \mid u_1, u_2)$$

XLM-R-based WiC model

- Fine-tuned XLM-R model for determining whether two usages of the same word express the **same sense**
- The predicted probability of **same sense** provides a continuous measure of semantic proximity
- Used as the **edge weight** between usages in the WUG

(Arefyev et al., 2021)

Datasets

Dataset	ID	Words	C1	C2
German	DWUG DE	24	1800–1899	1946–1990
Spanish	DWUG ES	60	1810–1906	1994–2020
English	DWUG EN	37	1810–1860	1960–2010

Three languages with WUG-based sense-level annotations

Experimental Setup

Usage pairs $\xrightarrow{\text{WiC model}}$ Semantic proximity

Semantic proximity $\xrightarrow{\text{WUG construction}}$ $G = (U, E, w)$

G $\xrightarrow{\text{Clustering}}$ Induced word senses

- WiC predictions provide the **edge weights** of the WUG
- Clustering methods induce **word senses** from the usage graph

Experimental Setup



- **Clustering configurations**
 - Different clustering algorithms and hyperparameters
- **Optimization objectives**
 - WSI: Adjusted Rand Index (ARI)
 - LSCD: Spearman's ρ
- **5-fold cross-validation**
 - Optimize configurations separately for WSI and LSCD

Optimization Directions

WSI optimization → Evaluate on LSCD

LSCD optimization → Evaluate on WSI

Do configurations optimized for one task transfer to the other?

WSI Evaluation Metric

Adjusted Rand Index (ARI)

$$\text{ARI} = \frac{\text{RI} - \mathbb{E}[\text{RI}]}{\max(\text{RI}) - \mathbb{E}[\text{RI}]}$$

- **Measures agreement** between induced senses and **human-derived senses** obtained from semantic proximity annotations
- **RI**: proportion of usage pairs on which the two clusterings agree
- $\mathbb{E}[\text{RI}]$: expected agreement under random cluster assignments

$\text{ARI} = 1$	$\text{ARI} = 0$	$\text{ARI} < 0$
Perfect agreement	Chance agreement	Worse than chance

LSCD Evaluation Metric

Spearman's Rank Correlation (ρ)

$$\rho = 1 - \frac{6 \sum_{i=1}^n d_i^2}{n(n^2 - 1)}$$

- Measures **agreement** between the predicted ranking of target words and the **gold ranking** of semantic change
- Captures the strength of the **monotonic association** between the two rankings

$\rho = 1$	$\rho = 0$	$\rho = -1$
Perfect agreement	No monotonic association	Perfect inverse agreement

WUG Construction: Hyperparameter Search

Complete WUG $\longrightarrow w(u_i, u_j) \in [0, 1]$

Parameter	Values explored
Threshold t	Edge weights below t are removed
Binarize	True / False
Normalize	True / False
Disconnected edges	True / False
Quantile	$\{1, \dots, 10\}$
Fill diagonal	True / False

- A complete WUG is initially constructed from predicted semantic proximity scores

Clustering Methods: Hyperparameter Search

Clustering method	Parameter	Values explored
Chinese Whispers	weighting	{lin, top, log}
Correlation Clustering	threshold_cc	0.5
WSBM	distribution	{binomial, poisson, discrete, normal, exponential}
Spectral Clustering	cluster selection	{Silhouette, Calinski–Harabasz, Eigengap}

German Results: WSI Optimization Transfers Strongly

Clustering method	Optimized for WSI		Optimized for LSCD	
	ARI	LSCD ρ	ARI	LSCD ρ
Chinese Whispers	.622 $\pm$.130	.768 $\pm$.241	.602 $\pm$.078	.768 $\pm$.241
Correlation Clustering	.622 $\pm$.132	.873 $\pm$.110	.666 $\pm$.096	.715 $\pm$.207
WSBM	.710 $\pm$.068	.911 $\pm$.156	.613 $\pm$.158	.828 $\pm$.169
Spectral – Silhouette	.771 $\pm$.090	.873 $\pm$.141	.715 $\pm$.067	.835 $\pm$.130
Spectral – Calinski–Harabasz	.755 $\pm$.087	.834 $\pm$.077	.702 $\pm$.087	.774 $\pm$.111
Spectral – Eigengap	.633 $\pm$.144	.807 $\pm$.202	.426 $\pm$.208	.596 $\pm$.213

Bold: higher LSCD ρ for the same clustering method.

Spanish Results: WSI Optimization Mostly Transfers

Clustering method	Optimized for WSI		Optimized for LSCD	
	ARI	LSCD ρ	ARI	LSCD ρ
Chinese Whispers	.475 $\pm$.049	.590 $\pm$.260	.453 $\pm$.066	.498 $\pm$.254
Correlation Clustering	.412 $\pm$.113	.661 $\pm$.106	.411 $\pm$.115	.615 $\pm$.121
WSBM	.444 $\pm$.022	.553 $\pm$.186	.480 $\pm$.078	.602 $\pm$.243
Spectral – Silhouette	.351 $\pm$.108	.578 $\pm$.173	.371 $\pm$.112	.420 $\pm$.187
Spectral – Calinski–Harabasz	.363 $\pm$.097	.420 $\pm$.239	.316 $\pm$.113	.301 $\pm$.271
Spectral – Eigengap	.310 $\pm$.084	.595 $\pm$.152	.301 $\pm$.094	.567 $\pm$.189

Bold: higher LSCD ρ for the same clustering method.

English Results: WSI Optimization Mostly Transfers

Clustering method	Optimized for WSI		Optimized for LSCD	
	ARI	LSCD ρ	ARI	LSCD ρ
Chinese Whispers	.243 $\pm$.098	.700 $\pm$.209	.256 $\pm$.129	.598 $\pm$.198
Correlation Clustering	.192 $\pm$.104	.307 $\pm$.194	.080 $\pm$.076	.293 $\pm$.189
WSBM	.226 $\pm$.144	.562 $\pm$.297	.045 $\pm$.025	.555 $\pm$.214
Spectral – Silhouette	.279 $\pm$.095	.600 $\pm$.205	.088 $\pm$.034	.505 $\pm$.233
Spectral – Calinski–Harabasz	.106 $\pm$.027	.157 $\pm$.425	.059 $\pm$.042	.288 $\pm$.215
Spectral – Eigengap	.064 $\pm$.038	.395 $\pm$.207	.054 $\pm$.050	.126 $\pm$.232

Bold: higher LSCD ρ for the same clustering method.

APD vs. Clustering-Based Approaches

Dataset	Best WSI → LSCD	APD
German	.873 ± .141	.760 ± .121
Spanish	.590 ± .260	.653 ± .250
English	.600 ± .205	.638 ± .202

Clustering-based approaches outperform APD only on German.

Key Findings

① WSI optimization transfers to LSCD

- WSI-optimized configurations transfer to LSCD across German, Spanish, and English
- LSCD-optimized configurations do not consistently transfer back to WSI

② Clustering does not consistently outperform APD

- German: **Clustering** > **APD**
- Spanish and English: **APD** > **clustering**

Key Findings

Hyperparameter Effects Are Dataset- and Method-Dependent

- No single configuration was optimal across datasets and clustering methods
- Optimal settings varied depending on the **dataset** and **clustering method**

Study 2: Key Takeaways

- Accurate semantic proximity judgments are essential for LSCD
- Specialized WiC models can achieve strong performance
 - Careful optimization of WUG and clustering configurations is important
- However, specialized WiC models require task-specific training

Large Language Models

Large Language Models are pre-trained on massive text corpora, acquiring broad:

- Linguistic knowledge
- Contextual knowledge

(Liu et al., 2019; Brown et al., 2020)

Can this broad knowledge support semantic proximity judgments?

Why Large Language Models?

- **LLMs can perform a wide range of NLP tasks through natural-language prompting**
 - Without task-specific fine-tuning
- **Practical opportunity for LSCD**
 - Prompt LLMs to judge whether two usages of a word express the same meaning
 - Avoid training specialized WiC models for each task
 - Enable more flexible adaptation across datasets and languages

(Brown et al., 2020)

From Clustering to Semantic Proximity

- Study 2 showed that clustering configurations affect LSCD performance
- Clustering quality depends on the semantic proximity judgments used to construct the WUG
- Can LLMs generate these semantic proximity judgments?

Research Questions

Can Large Language Models compete with specialized models in LSCD?

- ① Can automatically optimized prompts outperform manually crafted prompts?**
- ② Can LLMs perform the Graded Change LSCD task effectively?**
Can they outperform state-of-the-art WiC-based models?
- ③ Can LLMs outperform state-of-the-art LSCD models at the annotation level?**

Hypotheses

- ① Automatically optimized prompts improve LLM-generated semantic proximity judgments and downstream LSCD performance.
- ② LLMs can generate semantic proximity judgments that are competitive with those produced by specialized WiC models.

Metrics

Annotation-Level Evaluation

- SPR_WiC
 - Spearman correlation between **human judgments** and **model judgments** for usage pairs

Task-Level Evaluation

- SPR_LSCD
 - Spearman correlation between **gold graded-change scores** and **model predictions**

Models

- **Large Language Models**

- Llama 3.1 (8B)
- Mixtral 8×7B
- Llama 3.3 (70B)

- **Specialized WiC Baselines**

- **DeepMistake** (Arefyev et al., 2021)
 - SOTA on the **Russian and Spanish** datasets
 - MCL, enMCL, and MCL→es
 - Fine-tuned on WiC datasets across multiple languages
- **XL-LEXEME** (Casotti et al., 2023)
 - SOTA on the **English, Swedish, and German** datasets

(Jiang et al., 2024; Arefyev et al., 2021; Casotti et al., 2023)

Clustering Methods

- **Spectral Clustering**
 - von Luxburg (2007)
- **Agglomerative Clustering**
 - Jain and Dubes (1988)
- **Weighted Stochastic Block Model (WSBM)**
 - Peixoto (2019)

Clustering Methods: Hyperparameter Search

Clustering method	Parameter	Values
WSBM	distribution	binomial, poisson, discrete, normal, exponential
Spectral Clustering	number of clusters	Silhouette, Calinski–Harabasz, Eigengap
Agglomerative Clustering		

Prompt Design and Optimization

- **Prompt optimization**
 - Starting from the optimized prompt of Yadav et al. (2024)
- **Spanish adaptation**
 - English prompt translated into Spanish
- **Semantic proximity examples**
 - DUREl examples added to represent the **four semantic proximity scores**

(Yadav et al., 2024; Schlechtweg et al., 2018)

Prompt Design and Optimization

- **Prompt optimization with DSPy**
 - Further optimized the prompts using the DSPy framework
- **Optimization data**
 - Stratified sample from the resampled English dataset
 - Development set of the Spanish dataset
- **Objective**
 - Improve semantic proximity judgments for usage pairs

(Khattab et al., 2023)

Prompt Optimization Results

Dataset – Prompt	Llama 3.1		Mixtral		Llama 3.3	
	NOP	OP	NOP	OP	NOP	OP
DWUG ES – PrS	26.8	29.5	32.6	31.22	32.33	39.77
DWUG ES – PrE	26.5	35.5	33.7	34.80	40.88	46.33
DWUG EN – PrS	25.75	31.0	32.8	40.75	37.25	45.5
DWUG EN – PrE	28.75	37.25	33.25	38.75	35.75	49.25

Optimized prompts generally improve semantic proximity accuracy.

NOP = non-optimized prompt

OP = optimized prompt

PrS = Spanish prompt

PrE = English